

Mohammed Farah

(614) 549-8314 | mohamedfarahp14@gmail.com | linkedin.com/in/mohammed-a-farah | mohammedfarah.com

EDUCATION

Duke University

B.S. in Engineering – Electrical and Computer Engineering & Cultural Anthropology (Double Major)

Durham, NC

May 2027

EXPERIENCE

Baini Technologies (HemoSave Device)

January 2024 – Present

Co-Founder / Product Design & Research

Durham, NC

- Addressed critical gaps in postpartum hemorrhage monitoring (70,000+ maternal deaths annually) by conducting user research with 20+ medical professionals, translating clinical needs into detailed technical product specifications and validation metrics.
- Analyzed existing blood loss measurement solutions—visual estimation (50% accurate), gravimetric (10+ min), and premium devices (\$2K+)—synthesizing competitive findings into a stakeholder-facing product differentiation strategy.
- Defined product requirements and testing protocols achieving 95%+ correlation accuracy, matching premium-device performance at <\$500 cost and <2 min measurement time while balancing technical feasibility, safety, and manufacturability.
- Awarded **1st place** at the Rice 360 Global Health Design Competition among 20+ international teams, recognized for technical rigor and real-world clinical impact.

HOTP

July – August 2025

Operations Intern

Columbus, OH

- Mapped end-to-end invoicing workflows through structured stakeholder interviews, identifying key process gaps and pain points to surface actionable recommendations that directly informed ERP system enhancements.
- Synthesized operational data and user feedback into iterative solution models, presenting findings to cross-functional stakeholders to drive process changes and align teams on system design decisions.
- Designed and implemented an automated invoice digitization workflow end-to-end, reducing manual processing time by 70% and recapturing 15+ hours per week across the operations team.

Duke University

June 2025 – Present

Tour Guide

Durham, NC

- Lead interactive campus tours for groups of 10–30 prospective students and families, tailoring communication style and content to diverse audiences including international applicants and first-generation college students.
- Serve as a frontline ambassador for Duke's academic programs, research opportunities, and campus culture, fielding in-depth technical questions about engineering, pre-med, and interdisciplinary tracks.
- Coordinate with the admissions office to align tour narratives with institutional priorities, ensuring consistent and accurate representation of Duke's academic mission and student experience.

PROJECTS

Robotics Kinematics Library | Python

October 2025 – Present

- Developing an open-source Python library addressing gaps in existing robotics kinematics tools, implementing Denavit-Hartenberg algorithms for multi-joint systems and balancing mathematical accuracy with accessibility for students and engineers.
- Designing an intuitive API that abstracts transformation matrices and Jacobian calculations, reducing kinematics analysis workflows from hours to minutes and lowering the barrier to entry for robotics development.

Santa/Grinch Bot | Python, OpenCV, ROS2

October – December 2025

- Designing a dual-mode autonomous robot integrating a switching mechanism and multi-sensor array; developing OpenCV computer vision algorithms for real-time object classification and mode-dependent manipulation.
- Conducting iterative testing and validation cycles, collecting and analyzing performance data to continuously optimize recognition accuracy and refine mechanical and software design decisions.

LEADERSHIP & ACTIVITIES

Black Muslim Coalition

August 2025 – Present

Secretary

- Manage internal communications and document decisions for a 58-member organization, ensuring organizational continuity and coordinating follow-through on action items across all initiatives.
- Collaborate with Triangle Region Black and Muslim student organizations to co-host cross-community events, including Black Thanksgiving and Iftar, fostering interfaith and intercultural engagement on campus.

Member: Theta Chi, National Society of Black Engineers (NSBE), Duke Muslim Students Association (MSA)

TECHNICAL SKILLS

Product & Analysis: User Research, Competitive Analysis, Requirements Definition, Data Analysis, Workflow Design, Process Automation, Testing & Validation, ERP Systems

Languages: Python, C/C++, Java

Tools & Frameworks: OpenCV, ROS2, MATLAB, Git, KiCad, PSpice, Microsoft Office

Professional: Project Management, Stakeholder Communication, Technical Documentation, Cross-functional Collaboration